LoRA: Low Rank Adaptation of Large Language Models

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Original Authors: Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021).

Github Repo

Introduction / Abstract

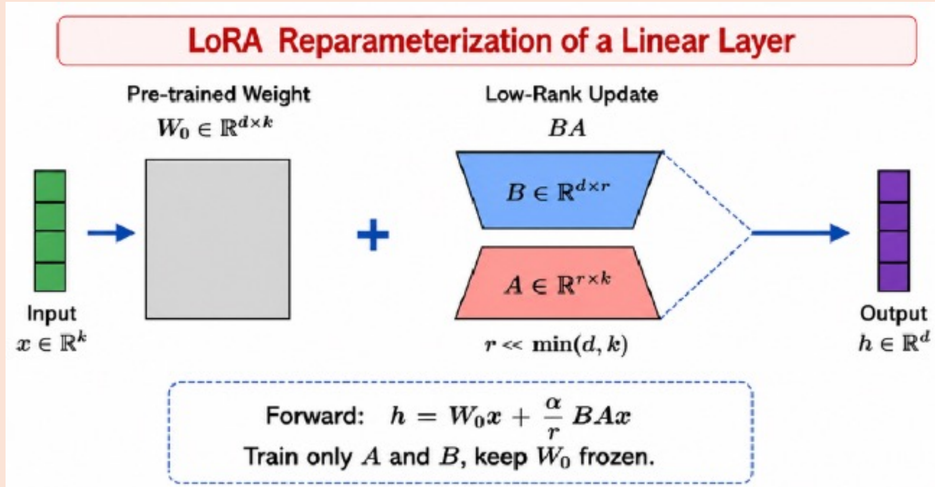
Problem: Full fine-tuning of large models is expensive in parameters, memory, and storage, making task-specific adaptation costly and inconvenient to deploy at scale.

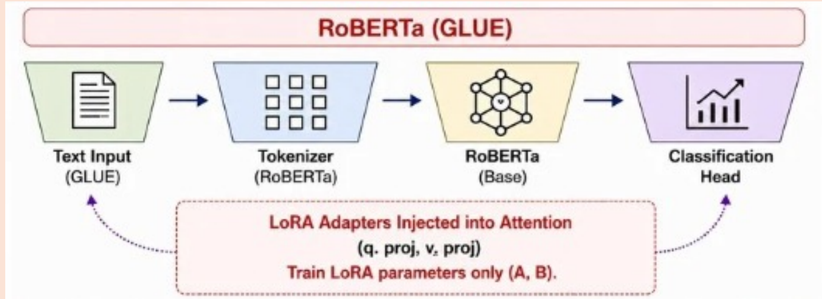
Goal: Reproduce the main claim of LoRA (Hu et al., 2021): low-rank adaptation achieves performance comparable to full fine-tuning while using far fewer trainable parameters.

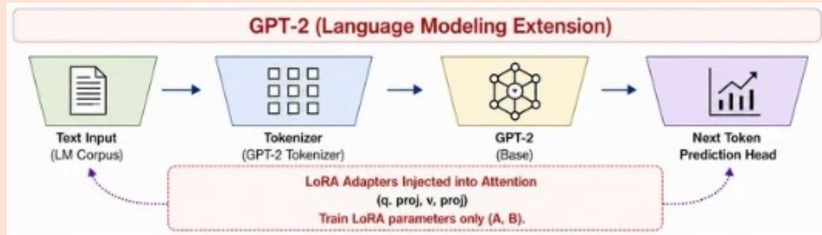
What We Reproduce/Investigate:

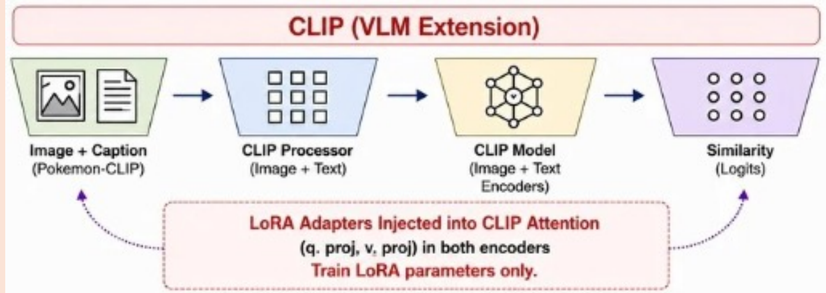
- Parameter-efficiency gains (trainable parameters, memory savings)
- Downstream task performance (RoBERTa on GLUE and GPT2-Medium on)
- Extension to vision-language models (CLIP) with image-text retrieval

Methodology









Results

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=== GLUE (RoBERTa-base, paper Table 2) ===
task  metric                 paper  ours  delta  wall_min
----  --------------------   -----  ----  -----  --------
MNLI  accuracy               87.5   87.9  +0.4   615.2
SST2  accuracy               95.1   94.8  -0.3   209.6
MRPC  accuracy               89.7   88.2  -1.5   24.8
COLA  matthews_correlation   63.4   62.3  -1.1   37.5
QNLI  accuracy               93.3   92.8  -0.5   138.5
QQP   accuracy               90.8   90.8  +0.0   489.2
RTE   accuracy               86.6   79.1  -7.5   44.8
STSB  pearson                91.5   90.9  -0.6   13.4
```

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=== E2E NLG (GPT-2 Medium, paper Table 3) ===
task  metric   paper  ours   delta   backend
----  -------  -----  -----  -----   --------
E2E   BLEU     70.40  69.78  -0.62   official
E2E   NIST     8.85   8.79   -0.06   official
E2E   METEOR   46.00  45.62  -0.38   official
E2E   ROUGE_L  71.80  71.29  -0.51   official
E2E   CIDEr    2.53   2.48   -0.05   official
```

Model		Total Params	Trainable Params
RoBERTa (base)	Full Fine-Tuning (FT)	~125 M	~125 M (100%)
	LoRA (r=8)	~125.3 M	**294,912** (~0.24%)
GPT-2 (medium)	Full Fine-Tuning (FT)	~355 M	~355 M (100%)
	LoRA (r=8)	~355.8 M	**786,432** (~0.22%)
CLIP ViT-B/32	Full Fine-Tuning (FT)	~151 M	~151 M (100%)
	LoRA (r=8)	~151.5 M	**491,520** (~0.33%)

LoRA reduces trainable parameters by ~99.7%+ across both NLP and VLM settings

query image

```
ground truth: a red cartoon character is flying through the air

Top retrieved captions from original CLIP:
1. a red and yellow pokemon pokemon character
2. a drawing of a blue and red pokemon
3. a blue and yellow pokemon character with its mouth open
4. a drawing of a cartoon character with arms and legs
5. a very cute looking pokemon character with big ears

Top retrieved captions from LoRA CLIP:
1. a red cartoon character is flying through the air
2. a cartoon character with a sword in his hand
3. a drawing of a cartoon character holding a wrench
4. a drawing of a cartoon character holding a surfboard
5. a drawing of a cartoon character with arms and legs
```

```
metric                 base     lora     delta
-----------------------------------------------
image_to_text_R@1      0.1750   0.5562   +0.3812
text_to_image_R@1      0.4250   0.5312   +0.1062
image_to_text_R@5      0.4563   0.8438   +0.3875
text_to_image_R@5      0.6750   0.8250   +0.1500
mean_recall@1          0.3000   0.5437   +0.2437
mean_recall@5          0.5656   0.8344   +0.2687
```

Conclusion

1. **Massive Parameter Efficiency:** We successfully reproduced the core claim of the original paper, achieving a ~99.7% reduction in trainable parameters across all models
2. **Uncompromised Performance:** Low-rank adaptation achieved comparable evaluation metrics to full fine-tuning across NLU (RoBERTa on GLUE) and NLG (GPT-2 on E2E)
3. **Successful Multimodal Extension:** Beyond traditional text models, we demonstrated the versatility of LoRA by successfully injecting adapters into CLIP's attention layers

Acknowledgements

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *arXiv*. https://doi.org/10.48550/arxiv.2106.09685

Novikova, J., Dušek, O., & Rieser, V. (2017). The E2E Dataset: New Challenges For End-to-End Generation. *Proceedings of the 18th Annual SIGdial Meeting on Discourse and Dialogue*, 201-206. https://doi.org/10.18653/v1/w17-5525

Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. (2018). GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding. *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*. https://doi.org/10.18653/v1/w18-5446